Seminar Thesis

Differentiable Bayesian Structure Learning

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Abstract

Causal structure learning (CSL) is a fundamental challenge across various scientific disciplines, aiming to uncover causal relationships between observed variables. Bayesian structure learning offers a probabilistic framework that infers the full posterior distribution over causal graphs, providing a quantification of the uncertainty in the causal discovery process. Differentiable Bayesian Structure Learning (DiBS) introduces a fully differentiable framework that leverages a latent continuous representation of discrete graph structures, enabling the application of gradient-based variational inference methods. DiBS is agnostic to local conditional dependency structures, making it broadly applicable even to complex, non-linear Bayesian networks. In two experiments we explore the limitations of DiBS. The first experiment demonstrates that the absence of prior knowledge about the causal graph structure can hinder accurate inference. The second experiment investigates the effects of sample size and finds that larger sample sizes do not necessarily lead to improved consistency or better performance of the DiBS algorithm.

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1 Introduction

Causal structure learning (CSL) is a fundamental problem in various scientific disciplines, aiming to uncover the underlying causal relationships between observed variables. Traditional causal discovery methods based on graphical models, such as the PC algorithm, Greedy Equivalence Search (GES), and Fast Causal Inference (FCI), primarily rely on statistical properties such as conditional independence to infer causal structures from purely observational data. Functional causal models (FCMs) provide an alternative approach by assuming specific functional forms for causal relationships, which, under suitable parametric assumptions, can identify the true causal graph rather than only its Markov equivalence class (MEC). However, these methods often face limitations in terms of scalability, interpretability and the ability to quantify uncertainty in causal inference (Glymour et al., 2019).

Bayesian causal structure learning has emerged as a powerful alternative to traditional approaches, offering a probabilistic framework to infer the marginal posterior distribution $P(G|D)$ over graph structures given the observed data. The underlying causal structures are often represented as directed acyclic graphs (DAGs) and modeled as Bayesian networks, where the primary challenges include the discrete and non-differentiable nature of the graph space, the enforcement of acyclicity constraints, and the combinatorial explosion of possible candidate structures. Despite these challenges, Bayesian approaches provide a natural way to quantify uncertainty, enabling researchers to obtain a full posterior distribution over possible causal graphs rather than a single point estimate or a Markov equivalence class. Uncertainty quantification is, however, not tied to the Bayesian setting: frequentist approaches such as the bootstrap also allow for it, as discussed below.

Several Bayesian CSL methods have been developed, including Markov Chain Monte Carlo (MCMC) approaches, which explore the posterior space by sampling in combinatorial spaces. However, MCMC methods often suffer from scalability issues due to the exponential search space. An alternative approach explored by Friedman et al. (1999) is the non-parametric DAG bootstrap, where classical causal discovery algorithms like PC or GES are applied multiple times on resampled data sets to approximate the distribution over graphs. Exact inference methods, while theoretically optimal, are usually infeasible in practice due to their computational complexity (Lorch et al., 2021).

The Differentiable Bayesian Structure Learning (DiBS) framework by Lorch et al. (2021) introduces a novel solution to these challenges by employing an elegant latent continuous representation of discrete graph structures. This approach enables the computation of gradients of the posterior distribution, facilitating the use of variational inference methods such as Stein Variational Gradient Descent (SVGD). A key contribution of DiBS is the integration of acyclicity constraints via a continuous latent prior, which allows for efficient gradient-based optimization. Furthermore, unlike many traditional Bayesian methods, DiBS does not require a conjugate setup, making it agnostic to the form of local conditional distributions. This flexibility allows DiBS to be applied across a wide range of Bayesian network models, including both linear and complex non-linear relationships,

e.g. modeled by neural networks.

The remainder of this seminar thesis is organized as follows:
In Chapter 2, we provide an in-depth overview of the methodological framework underlying Bayesian causal structure learning, discussing fundamental concepts such as Bayesian networks (Section 2.1) and the challenges associated with inferring the posterior distribution (Section 2.2). A key focus is placed on the differentiable formulation of Bayesian networks within a continuous latent space, which is explored in Section 2.3, with subsections detailing the bilinear generative model of directed graphs (Section 2.3.1) and the enforcement of acyclicity via latent priors (Section 2.3.2). Additionally, we examine the role of the Erdős-Rényi graph prior (Section 2.4) and posterior score estimation techniques (Section 2.5). In Section 2.6, we discuss the Stein Variational Gradient Descent (SVGD) algorithm for sampling from the posterior distribution using the score estimates.
Chapter 3 presents two experiments, exploring the importance of selecting an appropriate graph prior (Section 3.2), and the impact of sample sizes on the variability of edge inclusions (Section 3.3).
Finally, conclusions and future research directions are discussed in Chapter 4.

2 Method

Differentiable Bayesian Structure Learning (DiBS) is a causal discovery framework that uses observational data of a set of variables $X_1, \ldots, X_n$ to infer the underlying causal structure of these variables. The observations $D = \left\{x^{(1)}, \ldots, x^{(N)}\right\}$ are assumed to be independent from each other.

2.1 Bayesian Networks

The causal structure is modeled as a Bayesian network. Formally, a Bayesian network (G, Θ) is a probabilistic graphical model over a directed acyclic graph (DAG) G, in which every variable (node) depends only on the values of its direct predecessors (parents) and a stochastic noise term ε. Interpreting the edges of G as direct causal effects additionally requires causal assumptions such as causal sufficiency and faithfulness.
The existence of an edge in the DAG G from a parent node X_j to a child node X_i indicates that X_i is directly dependent on its parent X_j and conditionally independent of its non-descendants given all its parents. This leads to the factorization of the joint probability distribution in Equation (1), which makes the later computation of the likelihood efficient (Friedman et al., 1999):

$$P(X_1, \ldots, X_n) = \prod_{i=1}^{n} P(X_i | \mathrm{pa}(X_i)). \tag{1}$$

The set Θ describes the parametrization of the local conditional dependency structure. In a linear Gaussian Bayesian network, for example, the conditional dependency distributions take the form

$$X_i \sim \mathcal{N}\left(\sum_{j \in \mathrm{pa}(i)} \theta_{ij} X_j, \sigma_i^2\right), \tag{2}$$

with $\theta_{ij}, \sigma_i \in \Theta$. The DiBS framework, however, is not restricted to linear or Gaussian BNs and can be applied agnostic to the choice of local conditional distributions (Lorch et al., 2021).
The Bayesian network yields the following generative model with Θ depending on the graph G and the observational data D being generated jointly by G and Θ:

$$p(G, \Theta, D) = p(G)p(\Theta|G)p(D|G, \Theta). \tag{3}$$

Bayesian Inference of Bayesian Networks
The goal of DiBS is to not only estimate one possible Bayesian network, but to infer the marginal or joint posterior distribution $p(G|D)$ or $p(G, \Theta|D)$ of the Bayesian network from the observed data. The posterior distribution then allows us to make quantifiable statements about the uncertainty in the inference of the causal structure.
Bayes' theorem yields the decomposition of the posterior of a DAG into the (marginal) likelihood of the observed data given a graph G, the prior probability $p(G)$ encoding the initial beliefs about the graph structure in the numerator and the normalization constant $p(D)$ in the denominator:

$$p(G, \Theta|D) = \frac{p(D|G, \Theta)p(\Theta|G)p(G)}{p(D)}, \tag{4}$$

$$p(G|D) = \frac{p(D|G)p(G)}{p(D)}, \tag{5}$$

with $p(D|G) = \int p(D|G, \Theta)p(\Theta|G)\, d\Theta$.

2.2 Challenges in Inferring the Posterior

The inference of either posterior comes with certain challenges.
Firstly, the marginal likelihood $p(D|G)$ requires integrating over the conditional distribution parameters Θ, as shown in Equation (5). In most practical cases, this integral does not admit a closed-form solution, which also makes the marginal posterior intractable (Lorch et al., 2021).
The second challenge is the combinatorial complexity of the graph space. The number of possible directed acyclic graphs (DAGs) with d nodes grows super-exponentially as $O\left(d! \times 2^{\binom{d}{2}}\right)$, leading to an extremely large search space (Robinson, 1973). This exponential growth makes it practically infeasible to marginalize over all possible graphs even for a moderate number of variables. Thus, also the computation of the normalization constant given by

$$p(D) = \int \sum_{G} p(D|G, \Theta)p(\Theta|G)p(G)\, d\Theta, \tag{6}$$

becomes intractable.
The third major challenge for inference is the discrete nature of the graph structure in Bayesian networks. The discreteness of the posterior in G prevents the direct application of gradient-based inference methods.
Another key requirement of Bayesian network inference is to ensure that the inferred graph remains a directed acyclic graph (DAG). Enforcing this constraint introduces further complexity, as it necessitates specialized formulations that prevent cycles while allowing flexible exploration of the graph space.

2.3 Differentiable Formulation of Bayesian Networks in a Continuous Latent Space

Lorch et al. (2021) address these challenges by proposing a continuous relaxation of the discrete graph structure that allows for a lower dimensional and fully differentiable posterior inference.
Instead of sampling directly from the discrete distribution of a BN, Lorch et al. (2021) first translate the Bayesian network into a probabilistic continuous latent space. This is achieved by extending the generative model of the Bayesian network in Equation (3) by a continuous latent variable Z (see Figure 1) that assigns each edge in the directed

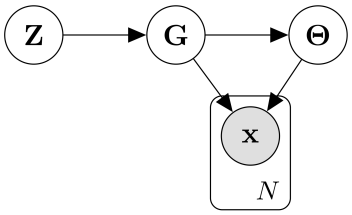


Figure 1: The generalized generative model of a Bayesian network with the additional latent variable Z. Figure taken from Lorch et al. (2021).

graph a continuous existence probability $p(Z_{ij}) \in [0, 1]$. The discrete graph G, represented by its adjacency matrix $G \in \{0, 1\}^{d\times d}$ now has a continuous twin $G_\alpha(Z) \in [0, 1]^{d\times d}$ that allows applying gradient-based Bayesian inference methods on the Bayesian network. The generalized generative model reads:

$$p(Z, G, \Theta, D) = p(Z)p(G|Z)p(\Theta|G)p(D|G, \Theta). \tag{7}$$

2.3.1 Bilinear Generative Model of Directed Graphs

Z itself is composed of two embedding matrices $U, V \in \mathbb{R}^{k\times d}$, with d being the number of variables and k being the latent dimension.
Lorch et al. (2021) choose to model each entry z_{ij} in Z as the scalar product of the vectors u_i and v_j. The vector u_i thus embeds the outgoing influence that node i has on all the other $d - 1$ nodes, while v_j embeds the effect of incoming edges on the value of node j. The edge probabilities are given by

$$p_\alpha(G|Z) = \prod_{i=1}^{d}\prod_{j\neq i}^{d} \sigma_\alpha\left(z_{ij}\right) = \prod_{i=1}^{d}\prod_{j\neq i}^{d} \sigma_\alpha\left(u_i^T v_j\right) \text{ with } \sigma_\alpha\left(x\right) = \frac{1}{1 + \exp\left(-\alpha x\right)}. \tag{8}$$

The sigmoid function $\sigma_\alpha\left(x\right)$ maps the scalar product to a valid probability in the interval $[0, 1]$. The inverse temperature α thereby acts as a hyperparameter in control of the *discreteness* of the outputted edge probability.
The resulting matrix of edge probabilities $G_\alpha(Z)$ is a relaxed representation of the discrete G matrix that converges to the discrete G for the inverse temperature α going to infinity:

$$G_\alpha(Z)_{ij} \xrightarrow{\alpha\to\infty} G_\infty(Z)_{ij} := \begin{cases} 1 & \text{if } \mathbf{u}_i^\top \mathbf{v}_j > 0 \text{ and } i \neq j, \\ 0 & \text{otherwise.} \end{cases} \tag{9}$$

Since we aim to model acyclic graphs that, in particular, do not contain self-loops, we set the diagonal edge probabilities to zero, $p_\alpha(g_{ij}|Z) := 0$.
To ensure that the model has sufficient capacity to capture complex dependencies between variables, the latent dimensionality should be set to $k \geq d$.

2.3.2 Enforcing Acyclicity via a Latent Prior

To overcome the challenge of acyclicity Lorch et al. (2021) make use of the differentiable characterization of acyclicity proposed by Zheng et al. (2018) and Yu et al. (2019), given

by

$$h(G) := \operatorname{tr}\left[(I + \frac{1}{d}G)^d\right] - d. \tag{10}$$

This term is exactly zero if and only if G is acyclic. The higher the degree of cyclicity of G, the larger the positive values $h(G) \geq 0$ takes. To avoid constrained optimization, this term is incorporated as a soft-constraint in the form of a prior on the latent variable Z. The prior is composed of independent Gaussians for the z_{ij} and a Gibbs distribution responsible for the penalization of the expected cyclicity of G given Z:

$$p_\beta(Z) \propto \exp\left(-\beta\ \mathbb{E}_{p(G|Z)}\left[h(G)\right]\right) \prod_{ij} \mathcal{N}\left(z_{ij}; 0, \sigma_z^2\right). \tag{11}$$

The inverse temperature β is a second hyperparameter to the DiBS framework, that controls the magnitude of the penalization of cycles in the graph. For $\beta = 0$ there is no penalization, while for very large values $\beta \to \infty$ the DAG acyclicity is enforced strictly. The choice of a soft-constraint simplifies the inference problem, but comes at the price that the inferred causal graph is not guaranteed to be cycle-free for $\beta < \infty$.

2.4 Erdős-Rényi Graph Prior

As DiBS is a Bayesian method, we have the flexibility to specify a prior term that incorporates our prior knowledge about the structure of the causal graph we aim to learn.
A common choice for a graph G with d nodes is the Erdős-Rényi prior given by

$$p(G) \propto q^{\|G\|_1}(1-q)^{\binom{d}{2}-\|G\|_1}. \tag{12}$$

The Erdős-Rényi prior gives every edge g_{ij} independently the identical prior existence probability q. The norm $\|G\|_1$ is equal to the number of active ($g_{ij} = 1$) edges in G and $\binom{d}{2}$ is the maximum number of edges that can be active in a DAG with d nodes, making $\binom{d}{2} - \|G\|_1$ the number of non-active edges. This prior formulation corresponds to setting the expected number of edges in the graph to $q\binom{d}{2}$ (Erdős and Rényi, 1960).
The influence of the choice of the hyperparameter q on the performance of DiBS is investigated in experiment 1 in Section 3.
Lorch et al. (2021) also implement DiBS with a scale-free graph prior and a uniform graph prior, which are not considered in this work. Either graph prior can be incorporated multiplicatively into the latent prior $p(Z)$ from Equation (11).

2.5 Posterior Score Estimation

Now, with the translation of the Bayesian causal structure learning problem into a continuous latent space, the gradient of the log posterior, called the latent posterior score, can be calculated with respect to the continuous Z instead of the discrete G. In the marginal case, the latent posterior score is given by

$$\nabla_Z \log p(Z|D) = \nabla_Z \log p(Z) + \frac{\nabla_Z \mathbb{E}_{p(G|Z)}\left[p(D|G)\right]}{\mathbb{E}_{p(G|Z)}\left[p(D|G)\right]}, \tag{13}$$

and for the joint posterior we get two gradients, one for the latent variable Z,

$$\nabla_Z \log p(Z, \Theta|D) = \nabla_Z \log p(Z) + \frac{\nabla_Z \mathbb{E}_{p(G|Z)}\left[p(\Theta, D|G)\right]}{\mathbb{E}_{p(G|Z)}\left[p(\Theta, D|G)\right]}, \tag{14}$$

and additionally the gradient with respect to the parameter set Θ,

$$\nabla_\Theta \log p(Z, \Theta|D) = \frac{\nabla_\Theta \mathbb{E}_{p(G|Z)}\left[p(\Theta, D|G)\right]}{\mathbb{E}_{p(G|Z)}\left[p(\Theta, D|G)\right]}. \tag{15}$$

Conveniently, the normalization constant is eliminated in the differentiation, as it does not depend on Z or Θ.
The derivation of the latent posterior scores is completely agnostic of the choice of local conditional distributions. However, depending on the distributional choice the expected likelihood under $p(G|Z)$ might not be differentiable (Lorch et al., 2021).

Gumbel-Softmax Trick
In the differentiable case, the Gumbel-*max* trick is applied. It states that any binary variable that follows a Bernoulli distribution $X \sim \text{Bernoulli}(q)$ can be rewritten using two independent Gumbel random variables $G_0, G_1 \sim \text{Gumbel}(0, 1)$:

$$X \overset{d}{=} \mathbb{1}\left[G_1 + \log q > G_0 + \log(1-q)\right]. \tag{16}$$

This property is now transferred to the inference problem to reparametrize G. Each edge g_{ij} is treated as the categorical random variable X and for q the probability $p_\alpha(g_{ij}|Z) = \sigma_\alpha(u_i^T v_j)$ from Equation (8) is plugged in.
Due to the unit step function, the expression is still not differentiable, which is why Jang et al. (2017) and Maddison et al. (2017) approximate it by the sigmoid function $\sigma_\tau(\cdot)$ with inverse temperature τ. For $\tau < \infty$ we obtain a soft relaxation of the unit step function, which is why this trick is called Gumbel-*softmax* trick. Within the sigmoid function the expression is rearranged to a " > 0"-inequality in Equation (18) and the difference of the two Gumbel random variables is summarized to a single logistic random variable $L = G_1 - G_0$. In Equations (20) to (22) the sigmoid function is replaced by its exponential definition and the log and exp are further simplified:

$$\begin{aligned}
g_{ij} &\overset{d}{=} \mathbb{1}\left[G_1 + \log p_\alpha(g_{ij}|Z) > G_0 + \log(1 - p_\alpha(g_{ij}|Z))\right] && (17)\\
&\approx \sigma_\tau\left(G_1 - G_0 + \log \sigma_\alpha(\mathbf{u}_i^\top \mathbf{v}_j) - \log(1 - \sigma_\alpha(\mathbf{u}_i^\top \mathbf{v}_j))\right) && (18)\\
&= \sigma_\tau\left(L + \log\left(\frac{\sigma_\alpha(\mathbf{u}_i^\top \mathbf{v}_j)}{\sigma_\alpha(-\mathbf{u}_i^\top \mathbf{v}_j)}\right)\right) && (19)\\
&= \sigma_\tau\left(L + \log\left(\frac{\exp(\alpha \mathbf{u}_i^\top \mathbf{v}_j)}{\exp(\alpha \mathbf{u}_i^\top \mathbf{v}_j) + 1} \cdot \frac{\exp(\alpha \mathbf{u}_i^\top \mathbf{v}_j) + 1}{1}\right)\right) && (20)\\
&= \sigma_\tau\left(L + \log\left(\exp(\alpha \mathbf{u}_i^\top \mathbf{v}_j)\right)\right) && (21)\\
&= \sigma_\tau\left(L + \alpha \mathbf{u}_i^\top \mathbf{v}_j\right). && (22)
\end{aligned}$$

All diagonal entries g_{ij} with $i = j$ are again set to 0. In the final expression the randomness in sampling $p(G|Z)$ stands separated from Z. The l_{ij} in Equation (23) represent the $d \times d$ independent logistic auxiliary random variables:

$$G_\tau(\mathbf{L}, \mathbf{Z})_{ij} := \begin{cases} \sigma_\tau\left(l_{ij} + \alpha \mathbf{u}_i^\top \mathbf{v}_j\right) & \text{if } i \neq j, \\ 0 & \text{if } i = j. \end{cases} \tag{23}$$

This handy property can now be exploited to calculate the likelihood gradient. The gradient ∇_Z is now independent from the expectation over $p(L)$. Thus, the gradient ∇_Z can be moved into the expectation and within the expectation the chain rule can be applied. Here it becomes apparent that the gradient of the likelihood with respect to G needs to be available:

$$\nabla_{\mathbf{Z}} \mathbb{E}_{p(G|\mathbf{Z})}\left[p(\mathcal{D} \mid G)\right] \approx \mathbb{E}_{p(\mathbf{L})}\left[\nabla_G p(\mathcal{D} \mid G)\big|_{G=G_\tau(\mathbf{L},\mathbf{Z})} \cdot \nabla_{\mathbf{Z}} G_\tau(\mathbf{L}, \mathbf{Z})\right]. \tag{24}$$

Score Function Estimator for the Likelihood Gradient
In the case of a non-differentiable likelihood, we can use the score function estimator by Williams (1992) to obtain an estimate for the gradient:

$$\begin{aligned} \nabla_{\mathbf{Z}} \mathbb{E}_{p(G|\mathbf{Z})}\left[p(\mathcal{D} \mid G)\right] &= \sum_{\mathbf{G}} p(\mathcal{D} \mid G) \nabla_{\mathbf{Z}} p(G \mid \mathbf{Z}) && (25) \\ &= \sum_{\mathbf{G}} p(\mathcal{D} \mid G) p(G \mid \mathbf{Z}) \nabla_{\mathbf{Z}} \log p(G \mid \mathbf{Z}) && (26) \\ &= \mathbb{E}_{p(G|\mathbf{Z})}\left[p(\mathcal{D} \mid G) \nabla_{\mathbf{Z}} \log p(G \mid \mathbf{Z})\right] && (27) \\ &= \mathbb{E}_{p(G|\mathbf{Z})}\left[\left(p(\mathcal{D} \mid G) - b\right) \nabla_{\mathbf{Z}} \log p(G \mid \mathbf{Z})\right]. && (28) \end{aligned}$$

The derivation of the score function estimator begins in Equation (25) by exploiting the linearity of the gradient operator, allowing the differentiation to pass inside the expectation. Next, in Equation (26), the log-derivative identity $\nabla_x \log f(x) = \nabla_x f(x)/f(x)$ is applied. In Equation (27) the definition of the expectation is used. Finally, Equation (28) introduces a constant b, which helps in reducing the variance of the gradient estimator. Since the expectation $\mathbb{E}_{p(\mathbf{G}|\mathbf{Z})}\left[\nabla_{\mathbf{Z}} \log p(\mathbf{G} \mid \mathbf{Z})\right]$ is zero, the subtraction of b does not affect the unbiasedness of the score function estimator (Mohamed et al., 2020). In practice, the expectation is estimated using Monte Carlo simulation, where multiple samples of G are drawn from $p(G|Z)$ to approximate the expectation empirically.
Both the Gumbel-softmax trick and the score function estimator can be applied for the joint likelihood $p(\Theta, D|G)$ in a similar fashion.

2.6 Stein Variational Gradient Descent

Lorch et al. (2021) now use the latent posterior scores with the Stein Variational Gradient Descent (SVGD) algorithm to generate a particle approximation of the graph posterior distribution $p(G|D)$. For only one particle, SVGD works exactly as a standard gradient ascent procedure, iteratively updating the current solution in the direction of the posterior score to eventually converge to a (local) maximum. This single maximum solution in the

case of one particle is equivalent to the maximum a posteriori point estimate (MAP) (Liu and Wang, 2016).
With DiBS, however, we want to generate samples from the whole posterior distribution rather than simply inferring the MAP estimate. Consequently, a set of M latent particles $\left\{Z_0^{(m)}\right\}_{m=1}^{M}$ is initialized and all particles are updated jointly in each SVGD step with step size η_t.
To prevent all particles from collapsing to the MAP solution, the update direction is given by ϕ_t instead of using the posterior score estimates directly:

$$\phi_t(\cdot) := \frac{1}{M} \sum_{k=1}^{M} \left[k(\mathbf{Z}_t^{(k)}, \cdot) \nabla_{\mathbf{Z}_t^{(k)}} \log p(\mathbf{Z}_t^{(k)} \mid \mathcal{D}) + \nabla_{\mathbf{Z}_t^{(k)}} k(\mathbf{Z}_t^{(k)}, \cdot) \right]. \tag{29}$$

The kernel $k((\mathbf{Z}, \mathbf{\Theta}), (\mathbf{Z}', \mathbf{\Theta}'))$ in $\phi_t(\cdot)$ measures how close the two particles Z and Z' and, in the joint case, also how close the parameters Θ and Θ' are. Lorch et al. (2021) use the following kernel:
With that, $\phi_t(\cdot)$ balances the search direction of each particle to point into a direction of high posterior probability $p(\mathbf{Z}_t^{(k)} \mid \mathcal{D})$, while not becoming too similar to all other particles. This allows the algorithm to explore the full posterior distribution.

$$k((\mathbf{Z}, \mathbf{\Theta}), (\mathbf{Z}', \mathbf{\Theta}')) := \exp\left(-\frac{1}{\gamma_Z} \|\mathbf{Z} - \mathbf{Z}'\|_2^2 \right) + \exp\left(-\frac{1}{\gamma_\Theta} \|\mathbf{\Theta} - \mathbf{\Theta}'\|_2^2 \right). \tag{30}$$

The hyperparameters α_t and β_t are annealed via a schedule that gradually increases them over the SVGD iterations. For $\alpha_t \to \infty$ this eventually leads to a single discrete graph structure for each particle. The annealing of $\beta_t \to \infty$ enforces the acyclicity of the resulting graph.
SVGD can be interpreted as a functional gradient descent that iteratively minimizes the KL divergence between the particle distribution and the true posterior distribution (Liu and Wang, 2016). The posterior over graphs is therefore represented by the resulting set of particles, which is essential for the DiBS framework.

Algorithm 1 DiBS with SVGD for inference of $p(\mathbf{G} \mid \mathcal{D})$

Input: Initial set of latent particles $\{\mathbf{Z}_0^{(m)}\}_{m=1}^{M}$, kernel k, schedules for α_t, β_t, and stepsizes η_t
Output: Set of discrete graph particles $\{\mathbf{G}^{(m)}\}_{m=1}^{M}$ approximating $p(\mathbf{G} \mid \mathcal{D})$

Incorporate prior belief of $p(\mathbf{G})$ into $p(\mathbf{Z})$
for iteration $t = 0$ to $T - 1$ **do**
    Estimate score $\nabla_{\mathbf{Z}} \log p(\mathbf{Z} \mid \mathcal{D})$ for each $\mathbf{Z}_t^{(m)}$
    for particle $m = 1$ to M **do**
        $\mathbf{Z}_{t+1}^{(m)} \leftarrow \mathbf{Z}_t^{(m)} + \eta_t \phi_t(\mathbf{Z}_t^{(m)})$
        where $\phi_t(\cdot) := \frac{1}{M} \sum_{k=1}^{M} \left[k(\mathbf{Z}_t^{(k)}, \cdot) \nabla_{\mathbf{Z}_t^{(k)}} \log p(\mathbf{Z}_t^{(k)} \mid \mathcal{D}) + \nabla_{\mathbf{Z}_t^{(k)}} k(\mathbf{Z}_t^{(k)}, \cdot) \right]$
    end for
end for
return $\{G_\infty(\mathbf{Z}_T^{(m)})\}_{m=1}^{M}$

3 Experiments

In this chapter, we present two experiments conducted on simulated datasets to further illustrate the working principles and limitations of DiBS. Both experiments have been implemented in Python using the official *dibs* package by Lorch et al. (2021)[1]. The experimental setup closely follows the configuration outlined in the original study by setting the number of nodes to $d = 20$, the latent dimensionality to $k = d = 20$, and running DiBS for 3000 SVGD iterations in the *dibs* package's standard settings. In both experiments, $M = 20$ particles are generated.
For simplicity, the experiments focus solely on inferring the marginal posterior $p(G|D)$, marginalizing over the parameters Θ. However, the experimental framework can be easily transferred to a joint posterior inference setup.

3.1 Evaluation Metrics

The performance of DiBS in both experiments is assessed using a holdout dataset of size $n_{ho} = 100$ and evaluated based on three metrics: Expected Structural Hamming Distance (E-SHD), Area Under the Receiver Operating Characteristic Curve (AUROC), and Negative Log-Likelihood (NegLL).

The **Expected Structural Hamming Distance (E-SHD)** measures the expected number of edge changes required to transform the inferred graph G into the ground truth graph G^*, considering their essential graphs that represent the Markov Equivalence Classes (MECs). It is computed by averaging the Structural Hamming Distance (SHD) over the posterior distribution of the inferred graphs. Lower E-SHD values indicate that the inferred structure is closer to the ground truth (Tsamardinos et al., 2006, Andersson et al., 1997).

$$\text{E-SHD}(p, \mathbf{G}^*) := \sum_{\mathbf{G}} p(\mathbf{G} \mid \mathcal{D}) \cdot \text{SHD}(\mathbf{G}, \mathbf{G}^*) \tag{31}$$

The **Area Under the Receiver Operating Characteristic Curve (AUROC)** is used to evaluate the quality of the inferred edge probabilities for pairwise edge predictions. It measures the ability of the model to distinguish between the presence and absence of edges by varying the decision threshold. The AUROC value ranges from 0 to 1, with a value of 0.5 indicating random guessing and a value close to 1.0 indicating excellent discrimination ability. The metric is computed under the inferred marginal posterior probabilities of the edge presence $p(g_{ij} = 1|D)$ (Friedman and Koller, 2000, Ellis and Wong, 2008).

The **Negative Log-Likelihood (NegLL)** evaluates the ability of the inferred graph to predict future observations from a held-out test set D_{test}. This metric penalizes models that assign low probability to test data under the learned distribution. Lower values of negative log-likelihood indicate better predictive performance (Murphy, 2001).

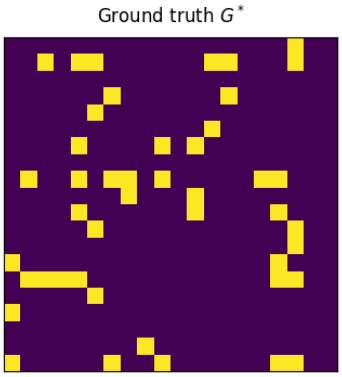
[1] `https://github.com/larslorch/dibs`

$$\text{neg. LL}(p, \mathcal{D}^{\text{test}}) := -\sum_{\mathbf{G},\Theta} p(\mathbf{G}, \Theta \mid \mathcal{D}) \cdot \log p(\mathcal{D}^{\text{test}} \mid \mathbf{G}, \Theta) \tag{32}$$

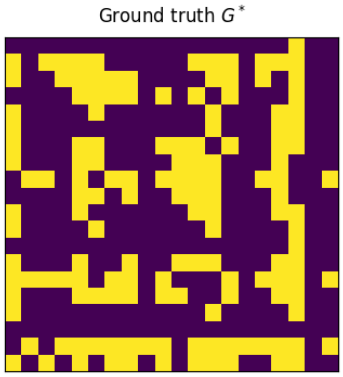
3.2 Experiment 1: The Importance of Selecting an Appropriate Graph Prior

Bayesian causal structure learning requires the specification of a suitable prior distribution $p(G)$ over the graph space to incorporate prior knowledge into the inference process. An appropriately chosen prior can guide the inference towards plausible solutions, potentially enhancing performance. However, in practical applications, such prior knowledge is often limited or entirely absent.
This experiment aims to investigate the impact of prior selection on the inference performance of DiBS, specifically addressing the question: How crucial is the choice of an appropriate graph prior, and to what extent does an incorrect prior affect performance?

Experimental Setup
To systematically analyze this aspect, we consider two configurations of the Erdős-Rényi (ER) graph prior, as introduced in Section 2.4. The first configuration ER(20) generates relatively sparse graphs with an individual edge existence probability of $q = 20\%$, while the second configuration ER(80) generates denser graphs in which each edge exists with a probability of $q = 80\%$. Applying the ER(20) prior corresponds to an expected number of $q\binom{d}{2} = 0.2 \times \binom{20}{2} = 38$ edges in the generated networks and the networks generated with the denser ER(80) prior contain $0.8 \times \binom{20}{2} = 152$ edges in the mean.



(a) Ground truth adjacency matrix G^* of a sparse ER(20) graph



(b) Ground truth adjacency matrix G^* of a dense ER(80) graph

Figure 2: Comparison of two randomly generated ground truth graphs using the Erdős-Rényi prior with different edge probabilities $q = \{0.2, 0.8\}$.

For each prior, we generate 10 ground truth graphs, each containing $N = 100$ observations. Subsequently, the DiBS algorithm (see Algorithm 1) is applied 10 times for each of the four experimental configurations outlined in Table 1, resulting in a total of 40 DiBS runs.

Results

Configuration	Ground Truth Data Model	DiBS Graph Prior
Configuration 1	ER(20)-graph	ER(20)-prior
Configuration 2	ER(20)-graph	ER(80)-prior
Configuration 3	ER(80)-graph	ER(20)-prior
Configuration 4	ER(80)-graph	ER(80)-prior

Table 1: Evaluated configurations of DiBS graph prior and ground truth data models in Experiment 1.

Figure 3 displays the resulting average metric values over the 10 DiBS runs for each of the four configurations from Table 1. A darker color indicates a better average score.

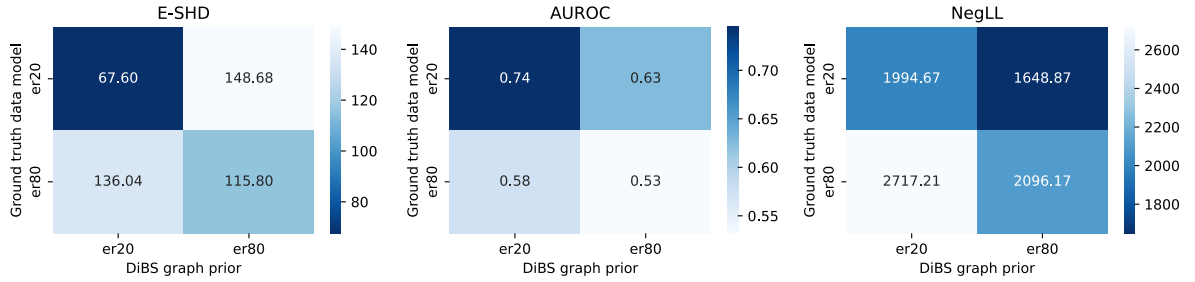


Figure 3: E-SHD, AUROC and NegLL for each of the 4 configurations averaged over 10 independent DiBS runs.

For the E-SHD the heatmap shows a clear picture: the lowest E-SHD (67.60 for ER(20) and 115.80 for ER(80)) is observed if in the DiBS inference the correct graph prior matching the ground truth is applied. Especially, for the sparse ER(20)-prior the matching prior leads to a significant reduction in E-SHD from 148.68 to 67.60, confirming that an appropriate prior leads to the most accurate structure recovery. A significant increase in E-SHD is observed when the DiBS prior mismatches the data generating prior, particularly in the case where the ER(80)-prior is applied to an ER(20) ground truth (148.68). This suggests that using an overly dense prior can introduce spurious edges, leading to a less accurate reconstruction.
The highest AUROC (0.74) is achieved when the ER(20)-prior is used for ER(20) data, meaning that this prior is well-suited for the given data and leads to the most reliable edge classification. When an ER(80)-prior is used for ER(20) ground truth, the AUROC drops to 0.63, indicating that the overly dense prior leads to an increase in false positives. For the more complex ER(80) ground truth, the AUROC values are lower across both priors, with the ER(20)-prior performing worse (0.58) compared to the ER(80)-prior (0.53). This suggests that DiBS with a sparse prior struggles to detect the true edge structure, whereas the dense prior might introduce noise, both resulting in poorer performance.
The NegLL also displays ambiguous results: The best NegLL performance (1648.87) is observed when the ER(80)-prior is used for the ER(20) ground truth. The use of the correct ER(20) prior in the inference process here led to an increase in the NegLL. For the

more complex ER(80) ground truth, the NegLL values are considerably higher, with the worst performance seen when using the ER(20)-prior (2717.21). Here, using the matching prior resulted in a substantial reduction in NegLL (2096.17), supporting the hypothesis that an appropriate prior improves inference outcomes.

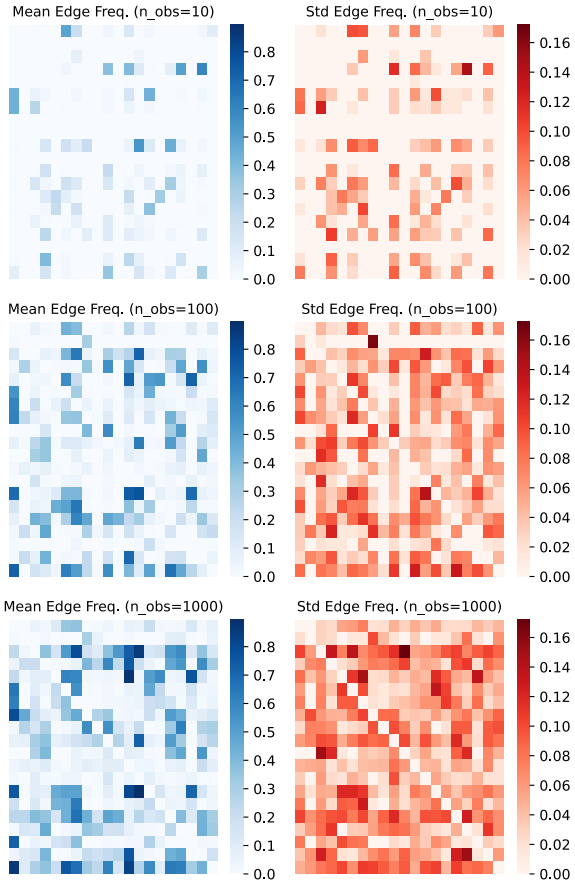
Generally, the results suggest that DiBS performs worse for ER(80) graphs compared to ER(20) graphs, which could be attributed to the increased complexity of inferring dense graph structures.
For the sparse ER(20) graphs, the use of the correct graph prior appears to contribute to improved inference performance in the E-SHD and the AUROC, aligning with theoretical expectations of Bayesian inference.
However, these experimental results do not provide conclusive evidence to fully support this claim. Several factors that are not regarded in this experiment may influence the observed outcomes, including the potential diminishing effect of the prior as the sample size increases. Additionally, the limited experiment size of 10 networks restricts the generalizability of the findings and may introduce random variability in the scores reported.

3.3 Experiment 2: Variability of Edge Inclusion for Different Sample Sizes

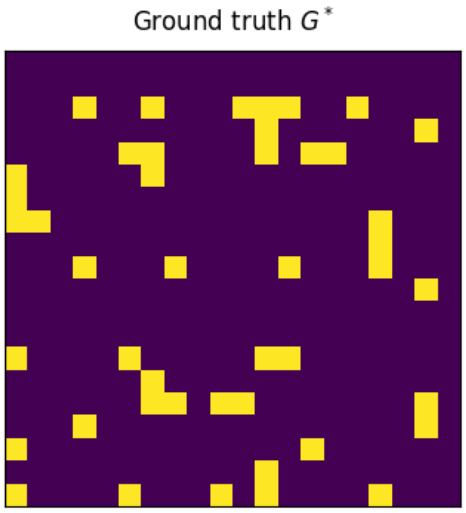
DiBS enables the analysis of uncertainty in the inference process by providing a full posterior distribution over possible Bayesian networks (BNs). In Experiment 2, we investigate the effect of sample size on the posterior variance of the causal structures inferred by DiBS. Directly measuring the variance of the inferred graph posterior is challenging, as the vast number of possible directed acyclic graphs (DAGs) results in minimal or no repetition of sampled graphs for $k = d = 20$. To address this, we focus on analyzing the variability in the inclusion of individual edges across the posterior particles, rather than considering entire graph structures.
The experiment aims to evaluate whether increasing the sample size leads to a reduction in posterior variance, resulting in more consistent edge selection and improved out-of-sample evaluation metrics.

Experimental Setup
We generated a single ground truth Bayesian network using the Erdős-Rényi graph model from Section 2.4 with default parameters, setting the expected number of edges per node to 2. Using this ground truth structure, we simulated three datasets with different sample sizes $N = \{10, 100, 1000\}$.
For each of the three datasets, we ran the DiBS framework 10 times, yielding a total of 30 independent runs. The ground truth data generating Erdős-Rényi graph model was passed as graph prior to every DiBS run. Within each DiBS run, $M = 20$ particles were sampled, producing an empirical distribution for the decision of including each edge in the graph. From these binary decisions, we calculated the edge inclusion probabilities (posterior edge frequencies) across the 20 particles for each run.
The mean edge frequencies were then computed by averaging the results across the 10 DiBS runs for each sample size. To further assess the variability of edge inclusion, the

standard deviation of the edge frequencies across the 10 DiBS runs is computed. The resulting mean and standard deviation matrices are visualized in Figure 4.



(a) The matrices of mean edge frequencies (left side in blue) and the standard deviations of edge frequencies (right side in red) for the sample sizes $N = \{10, 100, 1000\}$ calculated over 10 independent DiBS runs.



(b) The adjacency matrix of the Erdős-Rényi ground truth graph for experiment 2 with an expected number of two edges per node.

Figure 4: The ground truth graph (right) and the resulting mean and standard deviation matrices (left) for experiment 2.

Furthermore, the empirical distribution of the edge frequencies for all 400 edges across all 10 DiBS iterations are visualized in Figure 5 for each sample size.
Again, also the inference quality is evaluated by averaging the E-SHD, AUROC and NegLL over the 10 DiBS iterations for each sample size. The evaluation metrics are displayed in Figure 6.

Results

The mean matrices in Figure 4 show in combination with the histograms in Figure 5 that for $N = 10$ the majority of edge frequencies lie close to zero with only a few exceptions. This suggests that for very few observations like $N = 10$ DiBS is not able to capture

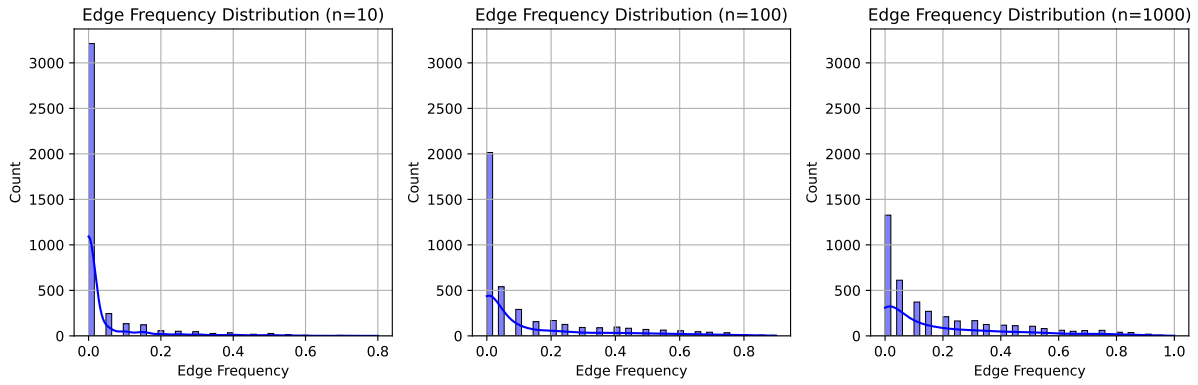


Figure 5: Empirical distribution of edge frequencies for all 400 edges across all 10 DiBS iterations for the sample sizes $N = \{10, 100, 1000\}$.

enough relevant causal patterns to learn an informative graph structure.
With access to more observations $N = \{100, 1000\}$ the edge probabilities seem to shift toward higher values, indicating that the confidence in the existence of certain edges increases. However, this shift can only be observed one-sided for the confidence in the existence of edges and not for the confidence of edges not existing. A general trend away from uninformative edge probabilities around 0.5 towards more extreme values close to 0 and 1 cannot be observed. This phenomenon could be explained by the sparse graph prior dominating the posterior distribution for small sample sizes and thereby enforcing edge probabilities close to 0. However, this explanation cannot be proven explicitly through this experimental setup.
Comparing the mean edge frequency matrices with the ground truth graph in Figure 4 supports the claim that the posterior distribution tends to become more informative for increasing sample sizes. It can be seen that especially the truly existing edges shown in the ground truth graph are assigned a higher posterior existence probability on average for $N = \{100, 1000\}$. The out-of-sample mean AUROC that increases from 0.64 for $N = 10$, over 0.92 for $N = 100$ to 0.96 for $N = 1000$ supports the hypothesis that a higher sample size leads to more precise posterior inference.
The observed trends in E-SHD and NegLL indicate a decline in inference performance with increasing sample sizes, which appears counterintuitive given the expectation that larger datasets should facilitate more accurate posterior inference. This observation may, however, be influenced by the limitations of the current experimental setup, which consists of only 10 DiBS iterations. The relatively small number of runs could introduce variability that affects the robustness of the reported results.

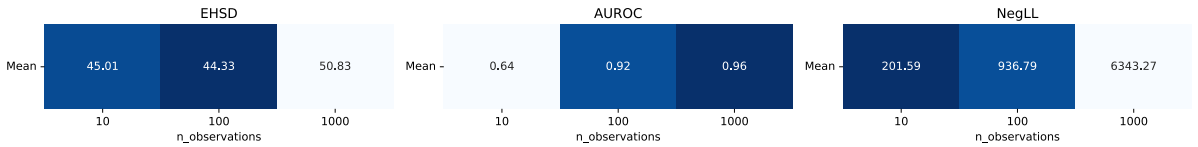


Figure 6: E-SHD, AUROC and NegLL for the sample sizes $N = \{10, 100, 1000\}$ averaged over 10 independent DiBS runs.

Moreover, the complexity of the underlying inference task with $d = 20$ may contribute to these findings. Despite an increase in the number of observations, the high-dimensional nature of the problem suggests that even a sample size of $N = 1000$ may remain insufficient to capture the full intricacies of the posterior distribution. The possibility that DiBS is affected by the curse of dimensionality cannot be ruled out, and may lead to challenges in distinguishing between genuine and spurious dependencies in the data.
Also, the standard deviations between the DiBS runs show that with more observations available, DiBS selects edges with overall less consistent frequencies. This could originate from the above explained effect of edge probabilities shifting away from zero for increasing sample sizes and thus generally exhibiting higher values.
Given these observations, and since the averages are reported without standard errors, the findings should be interpreted with caution, and further investigation is warranted to better understand the underlying causes of the observed trends.

4 Conclusion

The Differentiable Bayesian Structure Learning (DiBS) framework, introduced by Lorch et al. (2021), represents a significant advancement in causal discovery by addressing some of the key limitations inherent in traditional causal inference algorithms. Unlike classical approaches such as the PC algorithm, Fast Causal Inference, and Greedy Equivalence Search, DiBS provides a full Bayesian treatment of structure learning, representing the posterior over graphs by a set of particles and offering a quantification of uncertainty in the inferred causal relationships. This capability allows researchers to assess not only the most likely causal structures but also the degree of uncertainty associated with them, which is particularly valuable in high-dimensional and complex data settings.

A key strength of DiBS lies in its fully differentiable formulation, enabled by the use of techniques such as the differentiable latent acyclicity prior, the Gumbel-softmax trick, and score function estimators. This design makes DiBS compatible with gradient-based optimization techniques beyond the Stein Variational Gradient Descent (SVGD), broadening its applicability to other advanced learning methodologies. Additionally, DiBS is agnostic to the form of local conditional distributions, enabling its use with complex, non-linear Bayesian network models, thereby enhancing its flexibility across diverse domains.

Despite these advantages, practical challenges remain when applying DiBS to real-world causal discovery problems. One notable limitation, as highlighted by the results of experiment 1, is the sensitivity of DiBS to the choice of graph priors. In real-world applications where the true causal structure is unknown, the selection of an appropriate prior can be difficult, potentially necessitating larger sample sizes to achieve reliable inference results. Future research could further investigate this sensitivity and propose strategies for improving prior selection or adapting DiBS to scenarios with limited prior knowledge.

Experiment 2 demonstrated that even with the correct prior specified, causal discovery remains a challenging task due to its reliance on purely observational data. Support for interventional data was later added to the *dibs* package, which could make it an even more powerful tool for causal discovery. Future work could explore how this integration may further enhance inference results.

Another practical concern is the enforcement of acyclicity within the DiBS framework. Although the acyclicity constraint is managed through annealing of the hyperparameter β of the acyclicity prior, cyclic structures may still emerge among particles. The current approach of manually removing cyclic particles introduces the risk of altering the posterior distribution. More systematic solutions are needed to address this issue effectively.

Finally, similar to traditional causal inference algorithms, DiBS relies on several strong assumptions, including causal faithfulness and causal sufficiency. These assumptions can pose significant challenges in the interpretation of causal DAGs in real-world scenarios with non-simulated data. As such, caution should be exercised when analyzing the outcomes of DiBS-based inference, and additional robustness checks should be considered to mitigate potential misinterpretations.

In conclusion, the DiBS framework by Lorch et al. (2021) represents a major leap forward in causal discovery, offering a powerful and flexible tool that addresses key challenges such as uncertainty quantification and flexibility with respect to the local conditional distributions. However, its practical application to real-world causal inference remains

challenging, with issues such as sensitivity to graph priors, reliance on observational data, and strong causal assumptions requiring further research and refinement. These major challenges in causal discovery still prevail, and it is up to future research to overcome them.

A Electronic Appendix

The source code related to the experiments in Section 3 is available on GitHub:

https://github.com/adrian-io/diff-bayes-struct-learn

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Munich, 29.01.2025

Adrian Scholl